Wenjun CHENG

wjchengx10@outlook.com • 734-510-9639 • Ann

EDUCATION

University of Michigan (U-M) **Ann Arbor, MI**

- **B.S.E.** in **Robotics** (GPA: **4.0/4.0**) 08/2025 – 05/2027 (Expected)
- **Graduate-Level Coursework:** Foundation of Language Models (A), Mobile Robotics (A+), Multi-Robot Systems (A), Self Driving Cars (2026 Fall)
- **University Honors; Dean's List** 12/2025 & 04/2026
- **Michigan Engineering Continuing Student Scholarship** 08/2027

Shanghai Jiao Tong University (SJTU) **Shanghai, China**

- **B.S.E.** in **Electrical and Computer Engineering** (GPA: **3.65/4.0**) 08/2023 – 08/2027 (Expected)
- **Huatai Securities Technology Scholarship; Yu Liming Scholarship; Outstanding Undergraduate Scholarship** 11/2024

PUBLICATION

- HeRo-Nav: Heterogeneous Multi-Robot Collaboration for Semantic Navigation with Vision Language Models
  Siddhant Tandon, **Wenjun Cheng**, Junzhe Wu, Yue Hu, Yulun Tian
  Open World Navigation in the Foundation Model Era: Robustness and Failure Recovery, RSS 2026 Workshop, Accepted, 2026.

RESEARCH EXPERIENCE

Uncertainty-Aware Decentralized Multi-Robot Object Search **Ann Arbor, MI**
Undergraduate RA, advisor: Assistant Professor Yulun Tian (U-M Scalable Spatial Intelligence Laboratory) & Professor Vasileios Tzoumas (U-M Intelligent Robotics and Autonomy Laboratory) 06/2026 – Present

- Took charge of constructing simulation environments and evaluation scenarios for multi-robot exploration; Configured heterogeneous robot instances to validate semantic coordination under separate exploration demands.
- Refactored validated navigation modules from the HeRo-Nav project, including RGB-D occupancy mapping, frontier extraction, local path planning, and episode evaluation, to adapt to decentralized architecture requirements.
- Restructured the software architecture to eliminate global state synchronization assumptions; Implemented interfaces for pose exchange, task updates, frontier information, and compact semantic message exchange for decentralized operation.
- Perform benchmark experiments against centralized and communication-free baselines (ongoing).

Heterogeneous Multi-Robot Collaboration for Semantic Navigation with Vision-Language Models **Ann Arbor, MI**
*Undergraduate RA & **Co-first Author**, advisor: Assistant Professor Yulun Tian & Postdoc Yue Hu (U-M Scalable Spatial Intelligence Laboratory)* 02/2026 – Present

- Engaged in the complete research lifecycle to develop a zero-shot semantic navigation framework that enables coordinated task allocation among a heterogeneous multi-robot team.
- Independently constructed disaster-response simulation environments in NVIDIA Isaac Sim; Configured a quadruped, a humanoid, and a drone with different camera heights, base speeds, and terrain-dependent motion costs.
- Co-developed RGB-D to Bird's-Eye-View (BEV) mapping pipelines, including separated occupancy maps for each robot class, map fusion rules, persistent frontier identification, frontier-image caching, and candidate frontier filtering mechanisms.
- Collaboratively designed the multimodal prompt and structured output interface for the VLM coordinator; Implemented the integration of geometric planning and VLM baselines.
- Performed simulation experiments and analyzed benchmark results; HeRo-Nav achieved a 93.3% success rate (86.7% for the top classic baseline) and lowered the team hazard exposure from 40.1% to 25.7% across 15 hospital-disaster episodes; On the more difficult single-casualty episodes, it also outperformed the geometric baselines with an 88.9% success rate.
- Co-authored the manuscript which was accepted at two RSS workshops; Work on benchmark extension for future ICRA submission, designing more complex interdependent multi-robot collaboration scenarios (ongoing).

Probing Self-Motion Grounding in Vision-Language Models **Ann Arbor, MI** *Undergraduate RA & **Independent Project Lead**, advisor: Assistant Professor Yulun Tian & Postdoc Yue Hu (U-M Computational Autonomy and Robotics Laboratory & Scalable Spatial Intelligence Laboratory)* 01/2026 – Present

- Designed the experimental paradigm to analyze how VLMs encode camera self-motion from paired static observations and how navigation post-training reshapes spatial representations for embodied tasks.
- Developed an Isaac Sim-based data generation pipeline to build a controlled inverse-dynamics dataset covering diverse camera translations and rotations.
- Implemented scene-held-out evaluation to prevent the probes from memorizing scene appearance.
- Built a single-pass inference pipeline to extract hidden representations throughout both the vision encoder and causal language model of Qwen3-VL for systematic representation probing.
- Developed three complementary analytical tools, including semantic motion classification probes, Lie-algebra continuous regression probes, and model decision analysis modules.
- Introduced control groups to counteract confounding factors, including translation–rotation difficulty, multiple-choice bias, scene leakage, feature dimensionality, and mismatched image pairs.
- Conduct comparisons among base, SFT, and RL-navigated model variants, and plan to include ScanNet as a real-image validation set (ongoing).

Structured Spatial Memory and World Modeling for Vision-and-Language Navigation **Ann Arbor, MI**
Undergraduate RA, advisor: Assistant Professor Maani Ghaffari & Postdoc Yue Hu (U-M Computational Autonomy and Robotics Laboratory) 10/2025 – 02/2026

- Assisted in building a VLM-based long-horizon VLN framework, implementing scene graph construction and structured semantic memory modules for spatial world representation.
- Tested supervised fine-tuning approaches for foundation models, and ran extensive navigation trials to dissect system failure modes.
- Verified that semantic memory alone fails to resolve multi-view spatial ambiguity, and pinpointed weak VLM motion grounding as the critical limitation and inspiring follow-up probing research.

Development of a Multimodal Experimental Platform for Liquid-Impact and Fluid–Structure Interaction Analysis

Shanghai, China

*Undergraduate RA & **Project Leader**, advisor: Professor David L.S. Hung (SJTU Flow Diagnostics & Analysis Laboratory)* 04/2024 – 08/2025

- Led the design, assembly, and iterative optimization of a modular experimental platform for liquid-impact studies, incorporating vibration isolation and damping structures to mitigate structural resonance and improve measurement repeatability; Adopted by the lab as standard experimental infrastructure.
- Established a multimodal data acquisition pipeline integrating piezoelectric sensor, signal conditioner, oscilloscope, and high-speed imaging; Developed an infrared triggering circuit to achieve synchronized morphological and force measurements for micro-impact analysis.
- Designed controlled experiments with tunable parameters including release height, impact angle, needle size, and liquid type.
- Processed measurements via time-domain feature extraction and Fast Fourier Transform-based frequency analysis to separate genuine impact signals from structural vibration noise.
- Corroborated experimental measurements with theoretical impact models for platform validation.

Edge-Aware Active Perception and Exploration Planning for UAVs **Shenzhen, China**

Research Intern at Tsinghua University Shenzhen Graduate School 12/2023 – 02/2024

- Conducted literature review and comparative analysis of prevailing UAV path planning paradigms, and summarized key bottlenecks for edge-deployed active perception under computational constraints.
- Assisted in developing a hybrid exploration algorithm to balance exploration efficacy and computational cost for resource-limited aerial robots.
- Supported sim-to-real transfer and physical validation, deploying and testing planning algorithms on real UAV platforms to evaluate practical navigation performance.

PROFESSIONAL EXPERIENCE

Edan Instruments, Inc. **Shenzhen, China**

Algorithm Intern, R&D System Department 01/2025

- Prototyped the company's first interactive 3D heart visualization system for electrocardiographic (ECG) diagnostic results, translating abstract pathological diagnostic results into anatomically-accurate 3D visual presentations; Secured management approval for production integration.
- Established systematic mapping rules between ECG diagnostic labels and cardiac anatomical regions, and designed customized color rendering, regional highlighting, and pathological annotation strategies for different clinical findings.
- Processed and optimized the 3D heart model using Blender; Implemented and iterated core interactive functions (model rotation, dragging, and partial structure hiding) and refined rendering logic for stable and interpretable visualization.

TEACHING EXPERIENCE

SJTU Global College **Shanghai, China**

Tech Teaching Assistant (TA) for ENGR1000J Introduction to Engineering 09/2024 – 12/2024

- Mentored freshman engineering teams (about 40 students) through the full mechatronics design lifecycle, delivering hands-on technical instruction in C++ programming, SolidWorks modeling, and Arduino hardware prototyping.
- Co-designed innovative engineering projects; Received nomination for the **Annual Outstanding TA Award**.

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

SJTU Global College Tennis Association **Shanghai, China**

Founding President 08/2024 – 08/2025

- Founded the first tennis club within the Global College, drafting official club bylaws, planning long-term development strategies, and hosting committee elections.
- Recruited and oversaw over 100 registered members, creating beginner tennis guidance materials, organizing regular training based on their technical foundation, and developing a digital platform connecting members for practice partnerships.
- Coordinated more than 10 intra- and inter-institute competitions as well as tournament viewing outings.

SJTU Student Union **Shanghai, China**

Campus Concert Performer & Event Organizer –

- Coordinated concert arrangements by scheduling performers, managing onstage sequences, and supervising on-site facilities, including stage lighting and display systems to ensure smooth event execution.
- Performed as a piano and clarinet instrumentalist in campus concerts.

SJTU Global College **Shanghai, China**

Campus Ambassador –

- Designed promotional posters and organized on-site recruitment booths at alma mater.
- Provided ongoing online Q&A sessions to assist prospective students with academic and collegiate inquiries.

Shenzhen Zero-One College (Shenzhen X-Institute) & Tsien Hsue-shen Elite Class in Mechanics **Shenzhen, China**

Participant –

- Led two independent STEM projects ("Structural Superslip – Chibi Plan" and "Three-Stage UAV Dragonfly for Mars Exploration"), receiving **Best Design Award** and **Most Valuable Research Award** and selecting as distinguished "**Zero-One Student**".
- Attended academic visits and technical exchanges at multiple national research centers, including the National Center for Protein Science and the National Astronaut Training Center.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, MATLAB, Verilog
- **Robotics:** ROS 2, SLAM, Habitat, Isaac Sim, Probabilistic Robotics
- **Machine Learning:** Vision-Language Models (VLMs), LLM Architectures, PyTorch
- **Tools:** Git, Linux, Docker, LaTeX, SolidWorks

ADDITIONAL INFORMATION

- **Hobbies:** **Piano** (Grade 8, Royal Academy of Music, UK, advanced amateur level), **Clarinet** (Level 9, Central Conservatory of Music, advanced amateur level)